



Essay – Code-Based Questions

Lecture 3: Code Review (General Code Rules)

(Dr. Ehab Essa Exact Exam Style)

? Question 1 – Don't Repeat Yourself (DRY)

```
public class MathUtil {  
    public static void main(String[] args) {  
        int x = 10;  
        int y = 20;  
  
        int sum1 = x + y;  
        System.out.println(sum1);  
  
        int sum2 = x + y;  
        System.out.println(sum2);  
    }  
}
```

Questions:

1. Is this code correct functionally?
2. Identify the violated coding rule.
3. Explain briefly why this rule is violated.



Model Answer:

1. Yes, the code is functionally correct.
2. Violated rule: **Don't Repeat Yourself (DRY)**

3. The same logic ($x + y$) is duplicated instead of being reused.
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? Question 2 – Magic Numbers

```
public class Discount {  
    public static double calculatePrice(double price) {  
        return price * 0.15;  
    }  
}
```

Questions:

1. Is the code acceptable according to general code rules?
2. What is the main rule violated?
3. Why is this considered a problem?

✓ Model Answer:

1. No.
 2. Violated rule: **Avoid magic numbers**
 3. The value `0.15` has no clear meaning and should be replaced with a named constant.
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? Question 3 – One Purpose for Each Variable

```
public class Example {  
    public static void main(String[] args) {  
        int x = 5;  
        x = x * 2;  
        System.out.println(x);  
    }  
}
```

Questions:

1. Is the code readable?
2. Which rule is violated?
3. Explain the issue.

 Model Answer:

1. No.
 2. Violated rule: **One purpose for each variable**
 3. Variable `x` is used without a clear purpose or meaning.
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Question 4 – Use Good Names

```
public class Test {  
    public static void main(String[] args) {  
        int a = 50;  
        int b = 10;  
        int c = a - b;  
        System.out.println(c);  
    }  
}
```

Questions:

1. Is the code correct?
2. Which code rule is violated?
3. How does this affect maintainability?

 Model Answer:

1. Yes, it is correct.
 2. Violated rule: **Use good names**
 3. Poor variable names reduce readability and understanding.
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? Question 5 – No Global Variables

```
public class Counter {  
    static int count;  
  
    public static void increment() {  
        count++;  
    }  
}
```

Questions:

1. Identify the violated coding rule.
2. Why is this rule important?

✓ Model Answer:

1. Violated rule: **No global variables**
 2. Global variables increase coupling and make code harder to maintain.
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? Question 6 – Return Results, Don't Print Them

```
public class Calculator {  
    public static void add(int x, int y) {  
        System.out.println(x + y);  
    }  
}
```

Questions:

1. Is this method well designed?
2. Which rule is violated?
3. What is the problem with this approach?

✓ Model Answer:

1. No.
 2. Violated rule: **Return results, don't print them**
 3. Printing inside the method reduces reusability and testability.
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? Question 7 – Fail Fast

```
public class Division {  
    public static int divide(int x, int y) {  
        return x / y;  
    }  
}
```

Questions:

1. Which coding rule is violated?
2. Explain the issue briefly.

✓ Model Answer:

1. Violated rule: **Fail fast**
2. The method does not check for invalid input such as division by zero.

? Question 8 – Comments Where Needed

```
public class Data {  
    public static int process(int x) {  
        return (x * x) + 1;  
    }  
}
```

Questions:

1. Identify the violated rule.
2. Why is this rule important?

✓ Model Answer:

1. Violated rule: **Comments where needed**
2. The logic is unclear and requires explanation for maintainability.

? Question 9 – Use Whitespace for Readability

```
public class Demo{  
public static void main(String[] args){  
int x=10;int y=20;  
System.out.println(x+y);  
}  
}
```

Questions:

1. Is the code readable?
2. Which rule is violated?

✓ Model Answer:

1. No.
 2. Violated rule: **Use whitespace for readability**
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Essay Question 2 – Code & Code Review Rules

Consider the following Java code:

```
public class Student {  
  
    static int x;  
  
    public static void main(String[] args) {  
  
        int a = 50;  
        int b = 50;  
  
        if (a > 0 && b > 0) {  
            System.out.println((a + b) * 0.1);  
        }  
  
        if (a > 0 && b > 0) {  
            System.out.println((a + b) * 0.1);  
        }  
  
        int c = 3;  
        if (a == c) {  
            System.out.println("Equal");  
        }  
    }  
}
```

Answer the following questions:

1. Is the code **well-written according to general code review rules**?
 2. Identify **FOUR violations** of coding rules from Lecture 3.
 3. For each violation, mention the **related code review principle**.
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✓ **Model Answer (What Dr. Ehab expects):**

1. **No, the code violates several general coding rules.**
2. **Violations and principles:**
 - **Repeated code**
→ Violates **DRY (Don't Repeat Yourself)**
 - **Magic number 0.1 and 3**
→ Violates **Avoid magic numbers**
 - **Poor variable names (a, b, c, x)**
→ Violates **Use good names** and **One purpose per variable**
 - **Global variable x used without reason**
→ Violates **No global variables**
 - **Printing results instead of returning them**
→ Violates **Return results, don't print them**
 - **No validation for invalid input values**
→ Violates **Fail fast**
 - **Lack of whitespace and structure**
→ Violates **Use whitespace for readability**

(Any **four** earn full marks.)